

PROJECT SYNOPSIS

Name: Chandhan M, Kavya M

USN: P19MT24S12602 , P19MT24S126050

College: East Point College of Higher Education

Course: Master of Computer Applications (MCA)

Semester: 2nd Year / 3rd Semester

Project Title: AI-Based Student Learning Path Recommendation System Using Machine Learning and Deep Learning

Introduction:

With the rapid growth of digital education platforms, the need for personalized learning has become increasingly important. Traditional learning systems follow a common curriculum for all students, which may not address individual strengths and weaknesses. This project focuses on developing an AI-based system that recommends customized learning paths based on student performance and learning behavior.

Problem Statement:

Students often face difficulty in selecting appropriate subjects and learning sequences according to their academic performance. Existing systems lack adaptability and personalization, resulting in inefficient learning outcomes.

Objectives:

- To analyze student academic data using data science techniques.
- To group students based on learning behavior using machine learning algorithms.
- To recommend personalized learning paths using artificial intelligence.
- To enhance learning efficiency and academic performance.

Methodology:

The system uses a student dataset containing marks, interest levels, and study time. Data preprocessing and exploratory data analysis are performed to understand student behavior. K-Means clustering is used to categorize students into beginner, intermediate, and advanced groups. An Artificial Neural Network (ANN) is designed to improve learning path recommendations.

Tools and Technologies:

Python, Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, and Visual Studio Code.

Expected Outcome:

The proposed system provides personalized learning path recommendations for students, helping them focus on suitable topics and improve their academic performance. This project demonstrates the practical application of AI in the education domain.